PLAN A SAFE EXPERIMENT

Name: ..... **Date:** .....

Read the method. In the table, write one hazard and one precaution for each step. Then answer the multiple-choice questions.

METHOD: SEPARATING SALT FROM ROCK SALT

1. Grind the rock salt to a powder in a mortar with a pestle.
2. Add the powder to 50 cm^3 of water in a beaker and stir until the salt has dissolved.
3. Filter the mixture through filter paper in a funnel into a clean beaker.
4. Pour the salt solution into an evaporating basin.
5. Heat the basin gently on a tripod and gauze over a Bunsen burner until about half the water has evaporated and crystals appear at the edge.
6. Turn off the burner and leave the basin to cool before moving it.

RISK ASSESSMENT

Step	Hazard (what could cause harm)	Precaution (what you will do about it)
1		
2		
3		
4		
5		
6		

HEATING SAFELY

1. Why should you stop heating before all the water has evaporated?
   a) To save gas b) Hot salt spits out of the basin c) The crystals dissolve again
2. Which flame should a Bunsen burner be on when you are not heating anything?
   a) The yellow safety flame b) A roaring blue flame c) It must be turned off completely
3. What should you wear for step 5?
   a) Goggles only b) Goggles and a lab coat or apron c) Nothing special; salt is safe
4. How can you tell whether a tripod is still hot?
   a) Touch it quickly with a fingertip b) You cannot tell by looking; treat it as hot
   c) It glows red while it is hot

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PLAN A SAFE EXPERIMENT ANSWER KEY

Name: ..... **Date:** .....

RISK ASSESSMENT

1 | Grit flying up; heavy pestle | Wear goggles; grind gently and hold the mortar steady on the bench

2 | Spilt water makes the floor slippery; glass beaker | Wipe spills at once; carry the beaker with two hands, away from the edge of the bench

3 | Funnel and beaker can tip over | Support the funnel in a stand or hold it; pour slowly

4 | Spilling solution over hands or bench | Pour slowly over the bench, not over yourself; wipe drips

5 | Open flame; hot tripod and gauze; hot salt solution spitting | Goggles on, hair tied back, heatproof mat under the tripod; heat gently and stop before all the water has gone; do not lean over the basin

6 | Basin, tripod and gauze stay hot for a long time | Leave for at least 10 minutes; move the basin with tongs; treat everything as hot until proven otherwise

HEATING SAFELY

1. b) Hot salt spits out of the basin 2. a) The yellow safety flame 3. b) Goggles and a lab coat or apron 4. b) You cannot tell by looking; treat it as hot

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